

### **Practice Problems for Midterm**

1. BRAC University is planning to improve campus transportation services and wants to better understand the commuting patterns of its students. To support this initiative, a survey was conducted among 500 randomly selected undergraduate students from different departments. Information was collected on daily commuting time to campus (in minutes), monthly transportation expenditure (in BDT), number of days attended on campus per week, primary mode of transportation (bus, rickshaw, bicycle, private vehicle, or walking), year of vehicle manufacture and level of satisfaction with the existing transportation facilities categorized as low, medium, or high. The survey results indicated that students spent an average of 35 minutes commuting to campus each day. The university administration is considering these findings while developing future transportation policies and student support services.
  - a) What is the population and the sample in this context? [1]
  - b) Identify the parameter of interest and the statistic reported in this study. [2]
  - c) What type of statistics is primarily used to report the average commuting time in this study? [1]
  - d) List the variables and identify the type of each variable (qualitative or quantitative). If quantitative, specify whether it is discrete or continuous. Mention the scale of measurement for each variable. [5]
  
2. Delta Pharmaceuticals Ltd. regularly evaluates the efficiency of its quality-control operations. To evaluate efficiency, completion times (in minutes) were recorded during a production cycle. The observed completion times are:  
  
38, 62, 54, 16, 43, 30, 58, 24, 45-R, 35, 12, 50, 28, 41, 23, 61, 32, 44, 18, 52, 26+Q, 36, 56, 20, 48, 42, 31, 34, 14, 40

**$R$  = average of even digits of your student ID (rounded to the next integer),  $Q$  = last digit of your student ID**

- a) Construct a grouped frequency distribution table using class intervals using 10-20, 20-30 and so on as class intervals. [3]
- b) On the same graph, draw a histogram and the corresponding frequency polygon for the grouped data. [3]
- c) Determine the percentage of analysts who completed the inspection in less than 40 minutes. [1]
- d) Using the ogive, estimate the percentage of analysts whose completion times were between 35 and 45 minutes. [3]

3. LABAID Specialized Hospital conducted a study on the waiting times (in minutes) of patients before receiving medical consultation in its outpatient department. The grouped frequency distribution of waiting times is given below.

|                        |       |       |       |       |       |       |
|------------------------|-------|-------|-------|-------|-------|-------|
| Waiting Time (Minutes) | 10–15 | 15–20 | 20–25 | 25–30 | 30–35 | 35–40 |
| No of patients         | 7+L   | 16    | 26+L  | 34    | 24–L  | 8     |

**$L$  = the average of all digits in your student ID (rounded to the next integer)**

- Obtain three different measures that summarize the center of the waiting-time distribution and comment on these measures. [5]
  - Construct an ogive for the waiting-time distribution and use it to estimate the median waiting time. Compare your answer with the median obtained in part (a). [3]
  - The hospital plans to prioritize investigations for patients whose waiting times fall within the longest 10% of waiting times. Determine the minimum waiting time required to belong to this group. [2]
4. A retail company recently launched a new marketing campaign and wishes to evaluate the sales performance of its sales representatives. As part of a routine performance review, management selected 20 sales representatives from different regions and recorded the total value of sales achieved by each representative during the previous month. The monthly sales figures (in thousand BDT) are given below:

45, 80, 62, 95, 55, 70+P, 48, 83, 60, 88+Q, 52, 65, 75, 57, 90, 78–Q, 68, 85, 50, 72, 58, 77, 63, 92, 54, 81, 66, 73, 87, 59, 71, 84, 53, 69, 79

**$P$  = the last digit of your student ID,  $Q$  = the average of even digits of your student ID (rounded to the next integer)**

- Determine quartile values of the monthly sales figures and interpret your findings. [3]
- Suppose one additional sales representative had achieved monthly sales of 250 thousand BDT, and this observation had been omitted from the original dataset. Find a suitable measure of central tendency that can handle this situation. [3]
- Calculate the 75th percentile ( $P_{75}$ ) and the seventh decile ( $D_7$ ) of the monthly sales figures. Which measure represents a higher sales threshold? Briefly explain your answer. [3]
- Determine the monthly sales figure below which approximately 23% of the sales representatives are found. Comment on your result. [2]

5. A team of researchers from Bangladesh Agricultural University (BAU) conducted a field experiment to compare the effectiveness of two fertilizers on crop yield. Crop yields (kg per plot) were recorded from plots treated with Fertilizer A and Fertilizer B under similar environmental conditions.

**Fertilizer A:**

56.0, 42.0, 68.0, 50.0, 62.0, 45.0, 72.0, 54.0, 60.0 +  $Q$ , 48.0, 66.0, 52.0, 70.0, 58.0, 64.0

**Fertilizer B:**

52.0, 38.0, 64.0, 45.0, 60.0, 35.0, 58.0, 42.0 +  $2Q$ , 50.0, 66.0, 48.0, 55.0, 85.0 –  $Q$ , 40.0, 62.0

$Q$  = the average of the odd digits of your student ID (rounded up to the nearest integer).

- Compare the two fertilizers using all appropriate descriptive statistical measures. [4]
  - Determine whether either fertilizer contains any outliers. Hence, draw the corresponding boxplots for the two datasets on a common horizontal axis. [5]
  - Based on the boxplots constructed in part (b), compare the center, spread, and distributional shape of the two samples. [3]
  - Calculate a measure of skewness for each fertilizer. Comment on the shape of each distribution and determine whether your conclusions agree with the corresponding boxplots. [3]
6. An e-commerce company monitored the order processing times (in minutes) of customer orders at two fulfillment centers (Center A and Center B) during a promotional campaign. To evaluate operational performance, the processing times were summarized in the following grouped frequency distribution. The company plans to standardize its order processing strategy by adopting the practices of the fulfillment center that provides more consistent processing times.

| Order Processing Time | 20-30 | 30-40 | 40-50 | 50-60   | 60-70 | 70-80  | 80-90 |
|-----------------------|-------|-------|-------|---------|-------|--------|-------|
| Center A              | 4     | 7     | 10    | 12+ $D$ | 10    | 7      | 4     |
| Center B              | 7     | 10    | 9     | 12- $D$ | 10    | 8+ $D$ | 8     |

$D$

= the average of last four digits of your student ID (rounded up to the nearest integer).

- Determine the most appropriate measure of dispersion of the order processing times for each fulfillment center. Comment on your findings. [4]

- b) Calculate the coefficient of skewness for each fulfillment center and comment on the shape of the distributions. [3]
- c) Using an appropriate statistical measure, determine which fulfillment center should be selected by the company and justify your recommendation. [3]
7. A cognitive science research team wants to study how daily coffee consumption affects reaction time (measured in milliseconds). It is generally believed that moderate caffeine intake improves alertness and reduces reaction time (i.e., faster responses). A study is conducted on 15 adults, and the following data are collected:

| SL# | Coffee Intake<br>(cups/day) | Reaction Time<br>(ms) |
|-----|-----------------------------|-----------------------|
| 1   | $1 + A$                     | $320 - 5A$            |
| 2   | 1.5                         | 310                   |
| 3   | 2.0                         | 295                   |
| 4   | 2.5                         | 280                   |
| 5   | $3 + A$                     | $270 - 4A$            |
| 6   | $4 - A$                     | $260 + 3A$            |
| 7   | 3.8                         | 240                   |
| 8   | 3.0                         | 265                   |
| 9   | 2.7                         | 275                   |
| 10  | $4.5 - A$                   | 230                   |
| 11  | 1.8                         | 300                   |
| 12  | 1.2                         | 315                   |
| 13  | 2.3                         | 285                   |
| 14  | 3.9                         | 245                   |
| 15  | 1.6                         | 305                   |

Here, **A** is the average of the first and last digit of your student ID, rounded to the nearest integer.

- Identify the dependent and independent variables. [1 Mark]
- Draw an appropriate plot and describe the relationship between coffee intake and reaction time, considering the direction of the relationship between coffee intake and reaction time. [4 Marks]
- Can you mention a limitation of the plot shown in (b)? To address this limitation, which statistical measure should be calculated? Compute that measures and interpret your findings. [5 Marks]
- Fit a simple linear regression model to predict reaction time based on coffee intake. Interpret the estimates of the regression coefficients in this context. [5 Marks]
- Predict the expected reaction time for someone who drinks 3 cups coffee per day. [1 Mark]
- Calculate the residual for an individual who drinks 2.7 cups of coffee per day. Comment on the direction of the residual. [2 Mark]
- What percentage of variation in reaction time can be explained by coffee intake according to the model obtained in (d)? [2 Marks]
- Draw an estimated regression line on the same scatter plot drawn in (b). [4 Marks]
- If researchers model reaction time considering two independent variables: sleeping hours

and coffee intake, then what type of regression analysis would be appropriate for this situation? [1 Mark]